Supplementary Information: Genome-scale context, stoichiometric refinement, and neuro-symbolic reasoning for proteome function annotation

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S1 Benchmark suite overview, Stage 0 calibration, and Suite 1 evaluations

Supplementary Table S1 summarizes the eight benchmark suites evaluated in the main manuscript, including their cohorts, Stage 0 evidence channel configurations, active pipeline stages, primary comparators, and canonical result tables in the repository. Supplementary Table S2 reports the ancestor-propagated evaluation across the 13-genome and 21-genome reference panels comparing GSPA against production annotation pipelines, individual Stage 0 channels, and genomic language models.

Table S1. Overview of the eight GSPA evaluation suites, cohorts, Stage 0 channel configurations, active pipeline stages, and canonical result tables.

Suite	Evaluation Scenario & Cohort	Proteomes / Proteins	Stage 0 Channels	Active Stages	Primary Comparators	Canonical Result Table(s)
Suite 1A	CAFA3→CAFA5 Temporal GO Holdout	16,125 evals (3,440 at < 30% ID)	5-Ch (DIAMOND, Pfam, InterPro, ESM-2, ProtT5)	0 + 1 + 2A	DIAMOND, Protein for, DeepGOCNN, DeepGOPlus, DeepGOZero	scenario1_cafa_temporal_holdout_real.tsv
Suite 1B	Whole-Genome Withheld CAFA Splits + CAFA6	6 genomes (20,697 ORFs, 6 splits); 769 CAFA6	8-Ch (Retrained BLAST, InterPro-LR, 16-k CNN, ESM-2 3B, C-HMCNN)	0 + 1 + 2A + 2B + 3	8 Retrained Baselines + GSPA-Seq (Ours)	cafa_whole_genome_time_and_homology_splits.tsv; suite1_cafa6_prospective_summary.tsv
Suite 1C	Post-Cutoff Swiss-Prot EC (Holdout-125)	125 enzymes (74 species, 293 pairs)	1-Ch or 3-Ch (CLEAN, DeepProZyme, EnzBERT)	0 + 1 + 2B	CLEAN, DeepProZyme, EnzBERT, METEOR v8	scenario1_swissprot_ec_holdout125_real.tsv
Suite 1D	95-Genome Bacterial EC Homology Masking	95 genomes (69,666 proteins; 4 ID tiers)	1-Ch or 3-Ch (CLEAN, DeepProZyme, EnzBERT)	0 + 1 + 2B	Standalone Predictors, METEOR v8	scenario1_95genomes_ec_homology_real.tsv
Suite 2A	9-Genome Stage 1/2A Ablation & Perturbation	9 genomes (81 runs; 900 injected claims)	3-Ch (DIAMOND, HMMER/Pfam, InterProScan)	0 + 1 + 2A	Stage 0 Noisy-OR, Individual Priors B_1–B_6	pillar1_2_ablation_summary.tsv
Suite 2B	13/21-Genome SOTA & 88-Proteome Multi-Kingdom	13, 21, and 88 proteomes (4 domains)	3-Ch or 7-Ch (Full Neural + Structural)	0 + 1 + 2A	NCBI PGAP, InterProScan, deepFRI, gLM, gLM2	pillar1_suite2_panel13_21_sota.tsv; multikingdom_summary_by_domain.tsv
Suite 2C	1,000-Genome Bacterial Population Audit	1,000 RefSeq bacterial genomes	3-Ch (DIAMOND, HMMER/Pfam, InterProScan)	0 + 1 + 2A	DeepGO-SE, SPROF-GO, DeepGraphGO, DeepFRI, TALE, InterProScan	pillar1_suite2_1000bacteria_sota.tsv
Suite 2D	6-Eukaryote Anatomical Bridge & Relaxation	6 species (6,155 enslog.; 7,655 stress)	7-Ch (Full Neural + Structural)	0 + 1 + 2A	Direct GO Taxon Rules, Stage 2A Term Deletion	pillar3_eukaryote_anatomy.tsv; pillar3_relax_vs_remove.tsv
Suites 3–5	Role-Stratified LRO & Heteromeric Complex LSO	256/414 (*E. coli*), 144 (*B. subtilis*); 175 LSO	Track A (No PLM) or Track B (+ ESM-2 Centroids)	0 + 1 + 2B + 3	Sequence-Only/METEOR, RLGC Alone, GRDT	pillar2_lro15_summary.tsv; pillar3_complex_lso.tsv
Suite 6	RB-TnSeq Fitness Enzymes (Price-149) & 108-Audit	148 proteins (22 hosts); 108 GEMs; 6 BiGG	1-Ch or 5-Ch (CLEAN, MAPred, DPZ, EnzBERT, GraphEC, TopEC)	0 + 1 + 2B	6 Standalone Tools, METEOR v8, DCGM, ECrecer	pillar2_price149_all6_real.tsv; pillar2_invariant_audit_summary.tsv
Suites 7–8	BacDive Carbon Growth & Diagnostic Enzymes	529 (2,332), 3,017 (17,496), 6,313 (50,075)	3-Model Pool (gapseq, ModelSEED, CarveMe)	0 + 1 + 2B	CarveMe, ModelSEED, gapseq, METEOR, Traitar, DIASPora	pillar3_bacdive_track1_summary.tsv; pillar3_bacdive_track2_summary.tsv
Stage 3	De-Identified 120-Anomaly & 50-LRO Benchmark	120 anomalies (7 taxa); 50 LRO gaps (3 seeds)	Post-Stage-2 Candidate Set	1 + 2 + 3	Threshold ≥ 0.35, 5-Fold CV LR/RF/GBDT/MLP, No-Guardrail	pillar4_agent_evaluation.tsv; pillar4_multiseed_summary.tsv

Table S2. Ancestor-propagated evaluation across the 13-genome prokaryotic panel (11 prokaryotes), 21-genome multi-kingdom panel, and 9-genome genomic language model subset. On the 13-genome prokaryotic panel, strict leaf-level weighted $F_{\max}$ reaches 0.7215 for GSPA Stage 1+2 vs. 0.4438 for NCBI PGAP, 0.4015 for InterProScan 5.72, and 0.1565 for metagenomic-deepFRI; on the 21-genome panel, EC $F_{\max}$ reaches 0.8860 for GSPA 7-Channel and 0.8830 for GSPA 3-Channel vs. 0.8550 for DIAMOND-BLASTp, 0.8530 for CLEAN, 0.3880 for ProteInfer, and 0.3120 for InterProScan 5.72.

Panel Cohort	Method / Channel	Genomes (n)	Micro $F_{\max}$	CAFA $F_{\max}$	$S_{\min}$	Runtime / Peak RAM
13-Genome Panel (Prok. 11)	metagenomic-deepFRI (2026)	11	0.6010	0.5940	95.20	38.0 s / 4,800 MB
	InterProScan 5.72	11	0.7410	0.6780	58.40	680.0 s / 8,200 MB
	NCBI PGAP (2021)	11	0.7820	0.7680	54.20	1,800.0 s / 16,000 MB
	GSPA Stage 1+2 (3-Channel Full)	11	0.8421	0.8605	21.40	5.4 s / 4,812 MB
21-Genome Panel (All 21)	FoldSeek + ProstT5 3Di (2024)	21	0.6840	0.6310	83.24	10,800.0 s / 14,000 MB
	InterProScan 5.72	21	0.7180	0.6490	95.93	720.0 s / 8,200 MB
	ESM2-DeepGOPlus (2024)	21	0.7420	0.6950	105.29	420.0 s / 6,400 MB
	ProteInfer (2023)	21	0.7950	0.7540	38.12	120.0 s / 5,800 MB
	DIAMOND-BLASTp	21	0.8400	0.7890	93.81	17.0 s / 2,100 MB
	ProtT5 k-NN (2021)	21	0.8530	0.8010	39.40	28.0 s / 6,200 MB
	ESM-2 k-NN (2023)	21	0.8580	0.8080	37.85	24.0 s / 5,800 MB
	GSPA Stage 1+2 (3-Channel Core)	21	0.8690	0.8220	36.14	6.1 s / 4,812 MB
	GSPA Stage 1+2 (7-Channel All-Pooled)	21	0.8705	0.8407	35.20	6.8 s / 4,812 MB
9-Genome Subset	gLM / gLM2 Genomic LM (2024)	9	0.7280	0.7610	37.40	95.0 s / 7,200 MB

S1.1 Stage 0 Platt calibration parameters and correlation groups

In Stage 0 (`ScoreCalibrator.groovy` and `LikelihoodModel.groovy`), raw predictor scores $s(c) \in [0, 1]$ are mapped via Platt scaling $\tilde{q}(c) = \text{clip}(\sigma_{\log}(a_e s(c) + b_e), 10^{-4}, 0.9999)$ and multiplied by channel reliability $r_e \in (0, 1]$. Default calibration slopes and intercepts (a_e, b_e) are $(7.0, -3.5)$ for sequence alignment (DIAMOND/BLAST), $(6.5, -2.9)$ for profile HMMs (HMMER/Pfam), $(6.0, -2.4)$ for InterProScan and KofamScan, $(6.8, -3.2)$ for ESM-2 and ProtT5 k-NN transfer, $(6.2, -2.8)$ for ProteInfer, DeepGOWeb, and DeepGO-SE, and $(6.4, -3.0)$ for FoldSeek and ProstT5 3Di structural transfer. Default channel reliability weights r_e (`EvidenceType.groovy`) are 0.85 for AMR domain predictors, 0.80 for sequence homology and orthology, 0.75 for profile HMMs and sequence domains, 0.70 for protein-sequence ML, PLM k-NN, and structural similarity, 0.65 for structure-based deep learning, 0.60 for metabolic pathway context and inferred context, 0.55 for genomic language models, and 0.45 for raw genomic neighborhood context. Claims are partitioned into 11 correlation groups in `LikelihoodModel.groovy`: homology, protein-sequence ML, structure, genomic context, genomic language models, inferred context, sequence motifs, localization, region, viral, and specialized domain predictors.

S2 Hyperparameter specification and Stage 1–2A implementation crosswalk

All 18 configurable pipeline hyperparameters across Stage 1 (13 configuration keys exposed by `IntegrateCommand.groovy` and consumed by `IterativeRefiner.groovy` and the Stage 1 prior modules, alongside the fixed candidate partner gate 0.08 in `CoherencePrior.groovy`, the fixed co-operonic complex partner bonus 0.95 in `GenomicContextPrior.groovy`, the Stage 2A ancestor relaxation factor 1.0 in `ConsistencyEnforcer.groovy`, and the Stage 2A obligate-complex singleton prune threshold 0.45 in `IntegrateCommand.groovy`) and Stage 2B (5 parameters in `stoichiometric_locus_refiner.py`, alongside the fixed elemental-imbalance penalty $w_{\text{imb}} = 100.0$) were calibrated on the *Escherichia coli* K-12 MG1655 development proteome and held frozen across all out-of-sample prokaryotic, archaeal, viral, and eukaryotic benchmarks. Supplementary Table S3 reports these 18 configurable hyperparameters, their configuration keys, source modules, roles in inference, and default values (where penalty parameters are stored as positive magnitudes and applied with negative sign in log-odds space).

S2.1 Temporal GO and EC holdouts, whole-genome CAFA splits, and 95-genome homology masking

Supplementary Table S4 reports both the 16,125-evaluation CAFA temporal holdout benchmark (3,440 at $< 30\%$ sequence identity) and the complete 130-row whole-genome withheld CAFA evaluation across two time-based splits (`WG_Time_Rollover`, `WG_Time_Withheld`) and four homology-based splits (`WG_Homology_le100`, `WG_Homology_le70`, `WG_Homology_le50`, `WG_Homology_le30`) across six reference genomes (20,697 coding sequences) and 23 evaluated methods and ablations (`cafa_whole_genome_time_and_homology_splits.tsv` and `cafa_whole_genome_per_species.tsv`).

Table S3. Complete inventory of the 18 configurable GSPA v2.1.0 hyperparameters across Stages 1 and 2B.

Stage / Module	Symbol	Parameter Key	Source Module	Role in Inference	Default Value
Stage 1 (B_1)	α_{ess}	`alpha_ess`	`EssentialityPrior.groovy`	Unfulfilled essential GO function log-odds boost	1.50
Stage 1 (B_2)	α_{coh}	`alpha_coh`	`CoherencePrior.groovy`	Active pathway coherence log-odds scale	1.00
Stage 1 (B_2)	w_{part}	`alpha_complex_boost`	`CoherencePrior.groovy`	Obligate heteromeric complex partner boost ($0.65\,w_{\text{part}} = 0.5525$ for $N_{\text{strong}} \geq 2$)	0.85
Stage 1 (B_2)	w_{sing}	`alpha_complex_penalty`	`CoherencePrior.groovy`	Isolated obligate heteromeric singleton penalty magnitude (applied as -0.75)	0.75
Stage 1 (B_3)	α_{cons}	`alpha_cons`	`ConsistencyPrior.groovy`	SAT4J taxonomic conflict penalty magnitude (applied as -0.68)	0.68
Stage 1 (B_4)	α_{anat}	`alpha_anat_penalty`	`AnatomicalPrior.groovy`	Taxon-forbidden Uberon/PO/CL structure penalty magnitude (applied as -0.68)	0.68
Stage 1 (B_4)	α_{boost}	`alpha_anat_boost`	`AnatomicalPrior.groovy`	Active-structure developmental rescue log-odds boost	0.65
Stage 1 (B_5)	α_{gap}	`alpha_gap`	`GapFillingPrior.groovy`	Metabolic pathway hole-filling log-odds boost	1.50
Stage 1 (B_6)	α_{ctx}	`alpha_ctx`	`GenomicContextPrior.groovy`	Co-operonic pathway enrichment log-odds scale	1.00
Stage 1 (B_6)	$\alpha_{\text{gap_ctx}}$	`alpha_gap_ctx`	`GenomicContextPrior.groovy`	Co-operonic metabolic gap multiplier on $\alpha_{\text{ctx}}\, c_O(W)$	1.50
Stage 1 (Iter)	γ_{iter}	`damping`	`IterativeRefiner.groovy`	Iterative log-odds fixed-point damping weight	0.50
Stage 1 (Iter)	$K_{\max}$	`max_iter`	`IterativeRefiner.groovy`	Maximum Stage 1 message-passing iterations	6
Stage 1 (Iter)	ϵ	`epsilon`	`IterativeRefiner.groovy`	Stage 1 probability convergence tolerance	0.005
Stage 2B (SLR)	τ_{core}	`tau_core`	`stoichiometric_locus_refiner.py`	Core reaction probability threshold $S_{\text{rxn}}(j)$	0.50
Stage 2B (SLR)	α_{bio}	`alpha_bio`	`stoichiometric_locus_refiner.py`	Minimum required fraction of $v_{\text{bio}}^{\max}$ in pFBA	0.10
Stage 2B (SLR)	η_{null}	`tau_null`	`stoichiometric_locus_refiner.py`	Unassigned-gene hypothesis $h_{\varnothing}$ prior mass	0.05
Stage 2B (SLR)	γ_{FBA}	`gamma_fba`	`stoichiometric_locus_refiner.py`	Backward locus responsibility log-odds scale	2.20
Stage 2B (SLR)	γ_{damp}	`gamma_damp`	`stoichiometric_locus_refiner.py`	Geometric damping decay across outer passes	0.60

Supplementary Table S5 reports the 16-row post-cutoff Swiss-Prot enzyme benchmark (Holdout-125, $n = 125$ proteins across 293 ground-truth protein–EC pairs), separating standalone per-protein baselines, single-predictor refinement, and three-method ensemble pooling. Supplementary Table S6 reports the 12-row enzyme evaluation across the 95-genome bacterial panel under four donor-database homology-masking tiers (unfiltered, $\leq 70\%$, $\leq 50\%$, and $\leq 30\%$ sequence identity), comparing unfiltered METEOR v8 Top-1 changes against GSPA's single-predictor stoichiometric and pathway acceptance filter and GSPA's three-channel consensus acceptance filter, alongside exact discordant protein counts (b_{01}, b_{10}) and two-sided exact McNemar p-values.

Table S4. CAFA temporal holdout (16,125 protein–aspect evaluations, including 3,440 at $< 30\%$ identity) and complete 23-method whole-genome withheld CAFA evaluation across two time-based splits (`WG_Time_Rollover`, `WG_Time_Withheld`) and four homology-based splits (`WG_Homology_le100`, `WG_Homology_le70`, `WG_Homology_le50`, `WG_Homology_le30`) across six reference genomes (20,697 ORFs).

Panel A: 16,125-Eval Temporal Holdout	MF All	MF $< 30\%$	BP All	BP $< 30\%$	CC All	CC $< 30\%$	Macro $F_{\max}$ All	Macro $F_{\max} < 30\%$	Runtime / Peak RAM
DIAMOND (BLAST)	0.6224	0.0000	0.4437	0.0000	0.5805	0.0000	0.5489	0.0000	4.2 s / 1,150 MB
DeepGOCNN	0.4280	0.4209	0.3431	0.3389	0.6409	0.6487	0.4707	0.4695	8.5 s / 2,400 MB
DeepGOPlus (CNN+BLAST)	0.6337	0.4207	0.4613	0.3364	0.6722	0.6468	0.5891	0.4680	12.7 s / 2,650 MB
DeepGOZero	0.6552	0.5006	0.4514	0.3614	0.6602	0.6327	0.5889	0.4982	9.4 s / 2,800 MB
InterPro MLP	0.6561	0.5066	0.4592	0.3707	0.6667	0.6384	0.5940	0.5052	14.2 s / 3,100 MB
DeepGOZero+BLAST	0.6676	0.4982	0.4816	0.3614	0.6668	0.6327	0.6053	0.4974	13.6 s / 2,950 MB
ProteInfer	0.6966	0.6810	0.6146	0.5948	0.5301	0.5181	0.6138	0.5980	38.4 s / 6,400 MB
Stage 0 Noisy-OR (Ensemble)	0.7159	0.6949	0.6387	0.6144	0.7241	0.7075	0.6929	0.6723	56.8 s / 6,400 MB
GSPA (Iter 1)	0.7163	0.6971	0.6391	0.6152	0.7257	0.7202	0.6937	0.6775	59.2 s / 6,450 MB
GSPA (Iter 2)	0.7164	0.6972	0.6365	0.6143	0.7259	0.7202	0.6929	0.6772	61.5 s / 6,450 MB
GSPA (Iter 3)	0.7165	0.6972	0.6332	0.6104	0.7259	0.7202	0.6919	0.6759	63.8 s / 6,450 MB

Panel B: Whole-Genome Splits (Mean f_w / $F_{\max}$ / V)	Rollover f_w ($F_{\max}$)	Rollover V	Withheld f_w ($F_{\max}$)	Withheld V	$\leq 100\%$ f_w ($F_{\max}$)	$\leq 70\%$ f_w ($F_{\max}$)	$\leq 50\%$ f_w ($F_{\max}$)	$\leq 30\%$ f_w ($F_{\max}$)	$\leq 30\%$ V
DeepFRI (Pretrained Official)	0.2415 (0.2879)	1,300	0.2415 (0.2879)	1,300	n/a	n/a	n/a	n/a	n/a
DeepGO-SE (Pretrained Official)	0.4527 (0.4855)	113	0.4527 (0.4855)	113	n/a	n/a	n/a	n/a	n/a
Naive Term-Frequency Prior (Retrained)	0.2938 (0.3349)	1,570	0.2779 (0.3349)	1,570	0.3126 (0.3537)	0.3126 (0.3537)	0.3081 (0.3537)	0.3039 (0.3477)	5,274
DeepGOCNN (16-Kernel 1D-CNN, Retrained)	0.2802 (0.3520)	656	0.2514 (0.3357)	1,406	0.2695 (0.3533)	0.2639 (0.3539)	0.2617 (0.3499)	0.2582 (0.3398)	4,226
DIAMOND BLAST-*k*NN (Filtered DB)	0.3753 (0.4064)	988	0.3316 (0.3690)	1,195	0.3522 (0.3937)	0.3454 (0.3878)	0.3001 (0.3455)	0.1163 (0.1367)	5,519
DeepFRI (GCN + LM, Retrained)	0.3842 (0.4620)	61	0.3274 (0.4216)	308	0.3372 (0.4279)	0.3205 (0.4085)	0.3325 (0.4125)	0.2702 (0.3681)	1,949
DeepGOPlus (16-Kernel CNN + BLAST-*k*NN, Retrained)	0.3939 (0.4590)	489	0.3605 (0.4332)	959	0.3874 (0.4582)	0.3805 (0.4549)	0.3569 (0.4339)	0.2631 (0.3444)	4,433
ESM-2 3B Embedding *k*-NN (goPredSim, Retrained)	0.4138 (0.4835)	445	0.3097 (0.3939)	864	0.3286 (0.4080)	0.2259 (0.3097)	0.2416 (0.3148)	0.2340 (0.3137)	7,719
DeepGO-SE (ESM-2 3B + ELEmbeddings, Retrained)	0.4270 (0.4984)	83	0.3646 (0.4413)	257	0.3691 (0.4525)	0.3674 (0.4519)	0.3625 (0.4479)	0.3282 (0.4033)	2,139
ESM-2 3B MLP (Flat BCE, Retrained)	0.4343 (0.4981)	357	0.3729 (0.4538)	620	0.3824 (0.4646)	0.3761 (0.4631)	0.3711 (0.4563)	0.3374 (0.4194)	1,066
InterPro Domain Logistic Regression (Retrained)	0.4477 (0.5017)	916	0.4319 (0.4852)	1,114	0.4484 (0.5072)	0.4479 (0.5059)	0.4468 (0.5041)	0.4388 (0.4978)	3,330
ESM-2 3B C-HMCNN (Hierarchical MCLoss, Retrained)	0.4533 (0.5186)	237	0.3932 (0.4655)	572	0.3959 (0.4727)	0.3971 (0.4804)	0.3737 (0.4617)	0.3509 (0.4191)	1,245
TALE+ (Transformer + GO Label Attn + DIAMOND, Retrained)	0.4548 (0.5202)	216	0.3959 (0.4744)	371	0.4235 (0.4929)	0.4251 (0.4942)	0.4035 (0.4792)	0.3442 (0.4198)	2,787
DeepGraphGO (InterPro + GraphGCN, Retrained)	0.4551 (0.5162)	79	0.3980 (0.4753)	200	0.4033 (0.4847)	0.4023 (0.4827)	0.3824 (0.4633)	0.3378 (0.4274)	376
DPFunc (Domain Cross-Attn + GCN, Retrained)	0.4633 (0.5228)	318	0.3917 (0.4666)	631	0.3943 (0.4754)	0.3910 (0.4703)	0.3886 (0.4682)	0.3311 (0.4162)	3,413
SPROF-GO (PLM + Hierarchy + Label Diffusion, Retrained)	0.4861 (0.5467)	99	0.3976 (0.4746)	655	0.4129 (0.4929)	0.4240 (0.4982)	0.4043 (0.4821)	0.3435 (0.4215)	1,063
GSPA Stage 0 (Per-Protein Noisy-OR)	0.4882 (0.5389)	609	0.4330 (0.4856)	1,034	0.4521 (0.5127)	0.4773 (0.5302)	0.4811 (0.5304)	0.4856 (0.5215)	2,488
GSPA-Seq (Ours, Per-Protein Integrator)	0.4908 (0.5452)	670	0.4352 (0.4854)	1,219	0.4546 (0.5121)	0.4766 (0.5317)	0.4834 (0.5331)	0.4895 (0.5249)	2,756
GSPA + Operon Genomic Context Only (B_6)	0.4943 (0.5469)	670	0.4356 (0.4857)	1,219	0.4555 (0.5131)	0.4776 (0.5308)	0.4856 (0.5311)	0.4963 (0.5265)	2,756
GSPA + Pathway & Complex Coherence Only (B_1, B_2)	0.4927 (0.5460)	670	0.4361 (0.4873)	1,219	0.4585 (0.5169)	0.4776 (0.5337)	0.4827 (0.5327)	0.4893 (0.5250)	2,756
GSPA + Metabolic Gap-Filling & FBA Only ($B_5 + 2B$)	0.4911 (0.5453)	670	0.4356 (0.4855)	1,219	0.4550 (0.5125)	0.4765 (0.5319)	0.4834 (0.5331)	0.4895 (0.5249)	2,756
GSPA + Taxon & Anatomical Constraints Only ($B_3, B_4 + 2A$)	0.4946 (0.5467)	0	0.4437 (0.4862)	0	0.4645 (0.5126)	0.4862 (0.5323)	0.4959 (0.5338)	0.4988 (0.5254)	0
GSPA Full Whole-Genome (Stages 0 + 1 + 2A + 2B)	0.4996 (0.5490)	0	0.4455 (0.4883)	0	0.4685 (0.5160)	0.4886 (0.5323)	0.4968 (0.5314)	0.5055 (0.5272)	0

S3 Cross-ontology bridge and multi-kingdom population evaluations

Supplementary Table S7 summarizes the five resource tables programmatically extracted from the official OBO Foundry releases of the Gene Ontology (GO), Uberon multi-species anatomy ontology, Plant Ontology (PO), Cell Ontology (CL), and NCBI Taxonomy and loaded at runtime by the taxonomic and anatomical constraint modules. Supplementary Table S8 reports the 1,000-genome RefSeq bacterial population audit, and Supplementary Table S9 reports the 88-proteome multi-kingdom evaluation across Archaea, Bacteria, Eukaryota, and Viruses.

Table S5. Complete post-cutoff Swiss-Prot enzyme evaluation (Holdout-125, $n = 125$ proteins, 293 ground-truth EC annotations).

Predictor / Refinement Method	Masking Tier	Top-1 Acc. (Hits)	Top-3 Acc. (Hits)	Top-5 Acc. (Hits)	MRR	Micro $F_{\max}$
CLEAN (2023)	Unfiltered	32.8% (41/125)	40.8% (51/125)	41.6% (52/125)	0.3719	0.4360
DeepProZyme (2023)	Unfiltered	23.2% (29/125)	31.2% (39/125)	32.0% (40/125)	0.2750	0.2715
DeepProZyme + METEOR v8 (Single-Pred)	Unfiltered	25.6% (32/125)	31.2% (39/125)	31.2% (39/125)	0.2872	0.2804
DeepProZyme + GSPA Single-Pred (SLR + Coh.)	Unfiltered	23.2% (29/125)	31.2% (39/125)	32.8% (41/125)	0.2779	0.2703
EnzBERT (2023)	Unfiltered	32.8% (41/125)	37.6% (47/125)	39.2% (49/125)	0.3531	0.3596
Noisy-OR (3-Method Ensemble)	Unfiltered	33.6% (42/125)	40.8% (51/125)	43.2% (54/125)	0.3807	0.4381
GSPA (3-Method, Iter 1)	Unfiltered	32.0% (40/125)	42.4% (53/125)	44.0% (55/125)	0.3775	0.4035
GSPA (3-Method, Iter 2)	Unfiltered	32.0% (40/125)	42.4% (53/125)	44.0% (55/125)	0.3775	0.3891
GSPA (3-Method, Iter 3)	Unfiltered	32.0% (40/125)	43.2% (54/125)	44.0% (55/125)	0.3781	0.3830
EnzBERT	$\leq 70\%$ identity	30.4% (38/125)	35.2% (44/125)	35.2% (44/125)	0.3305	0.3204
EnzBERT	$\leq 50\%$ identity	28.0% (35/125)	33.6% (42/125)	33.6% (42/125)	0.3040	0.3163
EnzBERT	$\leq 30\%$ identity	17.6% (22/125)	24.8% (31/125)	24.8% (31/125)	0.2139	0.2212
CLEAN	$\leq 30\%$ identity	24.0% (30/125)	30.4% (38/125)	32.8% (41/125)	0.2741	0.2299
DeepProZyme	$\leq 70\%$ identity	23.2% (29/125)	31.2% (39/125)	32.0% (40/125)	0.2713	0.2715
DeepProZyme	$\leq 50\%$ identity	23.2% (29/125)	31.2% (39/125)	32.8% (41/125)	0.2720	0.2684
DeepProZyme	$\leq 30\%$ identity	22.4% (28/125)	31.2% (39/125)	32.8% (41/125)	0.2674	0.2650

Table S6. Single-predictor and 3-channel consensus acceptance filtering of candidate METEOR v8 EC changes across 95 bacterial genomes under four donor-database homology-masking tiers. p-values report two-sided exact binomial McNemar tests on discordant Top-1 predictions (b_{01}/b_{10}).

Predictor	Tier	Genomes / Proteins	Base Top-1 ($F_{\max}$)	METEOR v8 Top-1 (Δ)	METEOR b_{01}/b_{10} (p)	GSPA Single-Pred Top-1 (Δ)	Single-Pred b_{01}/b_{10} (p)	GSPA Multi-Pred Top-1 (Δ)	Multi-Pred b_{01}/b_{10} ($F_{\max}$)
CLEAN	Unfiltered	73/52,814	0.8214 (0.8796)	0.8234 (+108)	184/76 ($< 10^{-6}$)	0.8228 (+77)	99/22 ($< 10^{-6}$)	0.8243 (+155)	162/7 (0.8816)
CLEAN	$\leq 70\%$	95/69,663	0.8267 (0.8836)	0.8289 (+149)	191/42 ($< 10^{-6}$)	0.8283 (+108)	131/23 ($< 10^{-6}$)	0.8289 (+149)	154/5 (0.8851)
CLEAN	$\leq 50\%$	95/69,663	0.8271 (0.8844)	0.8294 (+162)	208/46 ($< 10^{-6}$)	0.8291 (+139)	156/17 ($< 10^{-6}$)	0.8293 (+155)	158/3 (0.8859)
CLEAN	$\leq 30\%$	94/68,681	0.8288 (0.8833)	0.8306 (+122)	156/34 ($< 10^{-6}$)	0.8302 (+93)	107/14 ($< 10^{-6}$)	0.8306 (+126)	132/6 (0.8845)
DeepProZyme	Unfiltered	95/69,666	0.9022 (0.9331)	0.9010 (−85)	103/188 (1.0×10^{-6})	0.9017 (−33)	24/57 (3.2×10^{-4})	0.9027 (+36)	51/15 (0.9335)
DeepProZyme	$\leq 70\%$	95/69,666	0.8965 (0.9290)	0.8961 (−27)	151/178 (0.152)	0.8963 (−15)	44/59 (0.167)	0.8975 (+67)	78/11 (0.9296)
DeepProZyme	$\leq 50\%$	95/69,666	0.8966 (0.9288)	0.8966 (+1)	163/162 (1.000)	0.8966 (−3)	49/52 (0.842)	0.8979 (+87)	93/6 (0.9297)
DeepProZyme	$\leq 30\%$	95/69,666	0.8967 (0.9289)	0.8970 (+19)	170/151 (0.315)	0.8966 (−4)	51/55 (0.771)	0.8976 (+67)	76/9 (0.9296)
EnzBERT	Unfiltered	95/69,666	0.8430 (0.8707)	0.8416 (−92)	100/192 ($< 10^{-6}$)	0.8428 (−10)	51/61 (0.395)	0.8438 (+59)	65/6 (0.8713)
EnzBERT	$\leq 70\%$	95/69,666	0.8394 (0.8675)	0.8374 (−146)	96/242 ($< 10^{-6}$)	0.8391 (−26)	41/67 (0.0157)	0.8405 (+70)	76/6 (0.8682)
EnzBERT	$\leq 50\%$	95/69,666	0.8330 (0.8602)	0.8304 (−181)	132/313 ($< 10^{-6}$)	0.8324 (−38)	31/69 (1.8×10^{-4})	0.8343 (+89)	93/4 (0.8611)
EnzBERT	$\leq 30\%$	95/69,666	0.8256 (0.8515)	0.8232 (−167)	108/275 ($< 10^{-6}$)	0.8254 (−18)	39/57 (0.082)	0.8267 (+78)	80/2 (0.8524)

Table S7. Summary of programmatic cross-ontology constraint tables in GSPA v2.1.0.

Resource File	Ontologies Bridged	Data Rows	Distinct Classes	Constraint Semantics & Examples
`go-taxon-constraints.tsv`	GO ↔ NCBITaxon	26,911	18,159 GO terms	13,819 `only_in_taxon` + 13,092 `never_in_taxon` rules (e.g., `GO:0009579` thylakoid)
`go-anatomical-constraints.tsv`	GO ↔ Uberon / PO / CL	2,715	2,674 GO; 1,294 Anat.	30 object properties (`results_in_development_of`, `occurs_in`, etc.)
`uberon-po-taxon-constraints.tsv`	Uberon / PO / CL ↔ NCBITaxon	8,331	1,247 Anat. classes	4,767 `only_in_taxon` + 3,564 `never_in_taxon` rules (e.g., `UBERON:0000948` heart → Metazoa)
`uberon-po-hierarchy.tsv`	Uberon / PO / CL DAG	7,722	2,855 Anat. classes	Transitive `is_a` and `part_of` edges across anatomical structures
`ncbi-taxon-hierarchy.tsv`	NCBI Taxonomy DAG	1,818	1,305 Taxon clades	Transitive taxonomic `is_a` lineage closure for benchmark clades

Table S8. Population-scale structural audit across 1,000 RefSeq bacterial genomes comparing deep-learning GO predictors, InterProScan, Stage 0 pooling, and full GSPA.

Method	Genomes (n)	Macro $F_{\max}$	Essential Core-22 (%)	Essential Full-29 (%)	Complex Coh. (%)	MetaCyc Comp. (%)	Taxon Viol. / Genome	Runtime / Peak RAM
DeepGO-SE / DeepGOMeta (2024)	1,000	0.5047	95.34%	82.17%	79.83%	11.59%	14,198.8	14.2 s / 4,200 MB
SPROF-GO (2023)	1,000	0.4930	99.91%	83.70%	79.18%	0.18%	7,322.7	28.5 s / 6,100 MB
DeepGraphGO (2021)	1,000	0.4020	99.28%	85.72%	70.56%	37.37%	9,922.2	19.4 s / 5,400 MB
DeepFRI (2021)	1,000	0.3843	100.00%	81.25%	98.87%	9.43%	106,815.2	24.0 s / 4,800 MB
TALE (2021)	1,000	0.3883	87.24%	75.38%	65.73%	19.63%	9,794.7	16.8 s / 4,500 MB
InterProScan 5	1,000	0.5940	97.36%	86.90%	61.89%	40.35%	940.4	145.0 s / 8,200 MB
GSPA Stage 0 (Noisy-OR)	1,000	0.6929	99.81%	90.42%	66.03%	36.71%	23,384.1	0.85 s / 3,400 MB
GSPA Stage 1+2 (Full)	1,000	0.6937	99.99%	95.29%	100.00%	66.75%	0.0	5.4 s / 4,812 MB

Table S9. Multi-kingdom proteome-wide evaluation across 88 proteomes (121,631 proteins) spanning Archaea, Bacteria, Eukaryota, and Viruses under homology masking.

Domain	Configuration	Genomes / Proteins	Mean $F_{\max}$ (All)	Mean $F_{\max}$ ($< 30\%$)	MF / BP / CC $F_{\max}$	Coverage	$\Delta F_{\max}$ [95% Bootstrap CI]	Runtime / Heap	Stage 3 Trigger (%)
Archaea	Stage 0 Noisy-OR	15/8,639	0.9125	0.5419	0.9460/0.9341/0.9146	0.9524	Baseline	0.35 s / 4,746 MB	1.97%
	GSPA Stage 1+2A	15/8,639	0.9129	0.5422	0.9454/**0.9345**/**0.9163**	0.9524	+0.0004 [−0.0001, +0.0010]	2.29 s / 5,120 MB	1.97%
Bacteria	Stage 0 Noisy-OR	47/104,032	0.7987	0.7987	0.8326/0.8304/0.8322	0.8443	Baseline	2.16 s / 4,479 MB	1.72%
	GSPA Stage 1+2A	47/104,032	0.8009	0.8009	0.8319/**0.8316**/**0.8371**	0.8443	+0.0022 [+0.0014, +0.0031]	12.66 s / 4,812 MB	1.72%
Eukaryota	Stage 0 Noisy-OR	6/6,155	0.7096	0.2459	0.8382/0.6709/0.7520	0.9079	Baseline	1.13 s / 5,676 MB	2.23%
	GSPA Stage 1+2A	6/6,155	0.7117	0.2453	0.8380/**0.6732**/**0.7555**	0.9079	+0.0021 [−0.0006, +0.0085]	6.67 s / 4,232 MB	2.23%
Viruses	Stage 0 Noisy-OR	20/2,805	0.7222	0.3112	0.8303/0.7726/0.7004	0.7576	Baseline	0.27 s / 4,841 MB	1.90%
	GSPA Stage 1+2A	20/2,805	0.7252	0.3116	**0.8304**/**0.7747**/**0.7037**	0.7576	+0.0030 [+0.0004, +0.0067]	0.42 s / 4,575 MB	1.90%

S4 Secondary metabolism, metagenomic dark matter, and Stage 2B stoichiometric refinement

In the Price-149 enzyme evaluation, the cognate-substrate stoichiometric neighborhood probe and outer-loop damped anchor are defined as follows. For each candidate pair (p, u), let $n_1(p, u)$ and $n_2(p, u)$ count chromosomal neighbors within ± 10 genes whose reactions share non-currency metabolites with u at 1 and 2 hops, $f_{\mathrm{op}}(p)$ denote the co-directional operon fraction, and $s_{\mathrm{nb}}(p, u) = \min(2.0, \sum_{|g'-g|\leq 10}(1 + 0.25|g' - g|)^{-1} P_{\mathrm{cand}}(g', u))$ measure distance-weighted neighbor EC support. The local stoichiometric probe shift $\Delta L_{\mathrm{probe}}(p, u) = 0.85 \min(1.5, 0.55n_1 + 0.20n_2) + 0.45 f_{\mathrm{op}}(p)\mathbb{I}[n_1 + n_2 > 0] + 0.35\mathbb{I}[u \in \mathcal{R}_{\mathrm{bal}}] - 0.50\mathbb{I}[u \notin \mathcal{R}_{\mathrm{ec}}]$ and Stage 1 neighbor coherence shift $\Delta L_{\mathrm{stage1}}(p, u) = 0.90 s_{\mathrm{nb}}(p, u) + 0.50 f_{\mathrm{op}}(p)\mathbb{I}[s_{\mathrm{nb}}(p, u) > 0.20] + 0.30\mathbb{I}[u \in \mathcal{W}_{\mathrm{kegg}}]$ are applied over $T = 3$ damped outer iterations ($\lambda_t \in \{0.70, 0.42, 0.252\}$):

$$P^{(t)}_{\mathrm{single}}(p, u) = \sigma_{\log}\Big(0.75 \operatorname{logit}\big(P^{(t-1)}(p, u)\big) + 0.25 \operatorname{logit}\big(P^{(0)}(p, u)\big) + \lambda_t\big(\Delta L_{\mathrm{stage1}}(p, u) + 0.65\, \Delta L_{\mathrm{probe}}(p, u)\big)\Big), \quad \text{(S1)}$$

and the multi-predictor posterior blends $0.55 P^{(t)}_{\mathrm{single}}(p, u) + 0.45 P_{\mathrm{NoisyOR}}(p, u)$. Supplementary Table S10 reports the biosynthetic gene cluster benchmark across 106 MIBiG/antiSMASH clusters, Supplementary Table S11 reports the 97-MAG uncultured microbial dark-matter panel across 16 phyla, and Supplementary Table S12 reports the 108-genome stoichiometric invariant audit, the comparison across six manually curated BiGG reference genome-scale metabolic models, and the role-stratified Price-149 breakdown on CLEAN.

Table S10. Biosynthetic gene cluster (BGC) recovery and non-cluster chromosomal specificity across 106 curated clusters from 12 bacterial species.

BGC Class	Clusters (n)	Core / Tailoring Loci	Stage 0 Core Rec.	GSPA Core Rec.	Stage 0 Tailoring Rec.	GSPA Tailoring Rec.	Non-BGC Specificity / Overall F_1 (Stage 0 → GSPA)
NRPS (Non-ribosomal peptide synthetase)	38	114/266	0.7807	0.9211	0.5414	0.8383	99.58% / 0.7420 → **0.8962**
Type I / II PKS (Polyketide synthase)	29	87/203	0.7586	0.9080	0.5222	0.8227	99.62% / 0.7227 → **0.8865**
Terpene / Carotenoid	18	36/90	0.8056	0.9444	0.5667	0.8556	99.69% / 0.7619 → **0.9136**
Siderophore (Aerobactin-like)	12	36/72	0.8333	0.9722	0.5972	0.8889	99.72% / 0.7935 → **0.9384**
RiPP / Bacteriocin / Lantipeptide	9	18/54	0.7222	0.8889	0.4815	0.7963	99.55% / 0.6842 → **0.8613**
All 106 BGCs	106	291/685	0.7801	0.9244	0.5387	0.8380	99.62% / 0.7395 → 0.8959

Table S11. Pathway gap resolution across 97 high-quality metagenome-assembled genomes (MAGs) of uncultured bacteria and archaea spanning 16 phyla.

Phylum / Group	Domain	MAGs (n)	Candidate Fills ($q \geq 0.30$)	High-Conf. Fills ($q \geq 0.50$)	Mean Fills / MAG ($q \geq 0.30/0.50$)	Pathway Comp. Before	Pathway Comp. After
Pseudomonadota	Bacteria	19	502	137	26.4/7.2	0.542	0.684
Actinomycetota	Bacteria	12	298	82	24.8/6.8	0.518	0.661
Bacteroidota	Bacteria	11	237	61	21.5/5.5	0.491	0.642
Bacillota	Bacteria	9	208	58	23.1/6.4	0.528	0.669
Cyanobacteriota	Bacteria	8	158	39	19.8/4.9	0.534	0.675
Chloroflexota	Bacteria	7	143	36	20.4/5.1	0.462	0.618
Halobacteriota	Archaea	7	132	32	18.9/4.6	0.489	0.631
Thermoproteota	Archaea	6	105	25	17.5/4.2	0.495	0.638
Other 8 Phyla (CPR, Asgard, etc.)	Mixed	18	392	98	21.8/5.4	0.469	0.613
All 97 Uncultured MAGs	**All (16 Phyla)**	97	2,175	568	22.4/5.9	0.511	0.656

Table S12. 108-genome stoichiometric invariant audit, curated BiGG reference GEM comparison ($n = 6$ species), and role-stratified Price-149 CLEAN refinement across Strata A, B, and C.

Panel A: 108-Genome Audit	Method / Reconstruction Pipeline	Active Rxns (LP Growth)	Flux-Carrying / Zero-Flux	Added Rxns (Cert. / Pad.)	Imbal. / Dead-End Active
108-Genome Audit	Raw Sequence Threshold (≥ 0.50)	2,809.5 (0% grow)	0.0/2,809.5	0.0 (0.0/0.0)	2,621.7/9.75
	METEOR v8 (Negative-Cost MILP)	3,121.0 (100% grow)	446.4/2,674.6	313.35 (51.01/262.34)	2,597.8/2.36
	Grow-Support Threshold Repair (Pos.-Cost LP)	2,584.8 (100% grow)	446.4/2,138.4	81.31 (81.31/0.00)	0.0/2.23
	GSPA Stage 2B (Lexicographic pFBA)	2,815.2 (100% **grow**)	300.6/2,514.6	5.70 (5.70/0.00)	0.0/0.00
Panel B: 6 Curated BiGG GEMs	**Reconstruction Pipeline**	**FBA Growing GEMs**	**EC Precision / Recall**	**EC F_1**	**BiGG Rxn Precision / Recall / F_1**
6 Curated BiGG GEMs	CarveMe v1.5.2 (Native BiGG Database)	6/6 (100%)	0.8217/0.8927	0.8536	0.6693/0.7305/0.6955
	Reconstructor (ModelSEED Namespace)	4/6 (67%)	0.4387/0.8787	0.5844	ModelSEED namespace (672.3 active rxns)
	DeepProZyme + METEOR v8 (Neg.-Cost MILP)	6/6 (100%)	0.4383/0.8842	0.5851	0.1655/0.2900/0.2091
Panel C: Price-149 Strata	**Progression: Base → Probe → 1-Pred → Multi-Pred**	**Reaction Role**	**Micro F_{max} Trajectory**	**Macro F_1 Trajectory**	**Top-1 Hits (Acc.) Trajectory**
Price-149 Stratum A ($n = 91$)	CLEAN Base → + Stage 2B Probe → + Single-Pred → + **Multi-Pred**	Core Biomass	0.6336 → 0.6405 → 0.6274 → **0.7019**	0.4740 → **0.5288**	46 → 46 → 47 → **52** (57.1%)
Price-149 Stratum B ($n = 55$)	CLEAN Base → + Stage 2B Probe → + Single-Pred → + **Multi-Pred**	Catabolic	0.3580 → 0.3753 → 0.3461 → **0.4546**	0.3103 → **0.3436**	13 → 13 → **15** (27.3%) → 13 (23.6%)
Price-149 Stratum C ($n = 2$)	CLEAN Base → + Stage 2B Probe → + Single-Pred → + **Multi-Pred**	Secondary	0.5099 → 0.5099 → 0.5175 → **0.5833**	0.5000 → 0.5000	1 → 1 → 1 → 1 (50.0%)

S5 Experimental phenotype prediction in BacDive and statistical testing

In the BacDive benchmark, the five-fold genus-blocked cross-validation assigns each taxonomic genus deterministically to one fold via MD5 hashing. Within each training fold, tool-specific conditional probabilities $p_{g,\mathrm{cal}}, p_{m,\mathrm{cal}}$, assay

prevalence $a_{\text{prev}}(y)$, and fold-local pooled probability $p_{\text{pool}}(s, y)$ ($0.38p_{g,\text{cal}} + 0.18p_{m,\text{cal}} + 0.22a_{\text{prev}} + 0.22P_{\text{Track1}}$ for carbon; $0.36p_{g,\text{cal}} + 0.16p_{m,\text{cal}} + 0.22a_{\text{prev}} + 0.26P_{\text{Track1}}$ for enzymes) are estimated strictly from the training split. Supplementary Table S13 reports the per-fold five-fold genus-blocked cross-validation metrics across both panels.

Table S13. Summary of 5-fold genus-blocked cross-validation in BacDive across carbon-source growth ($n = 2{,}332$ assays, 529 strains) and diagnostic enzymes ($n = 17{,}496$ assays, 3,017 type strains).

Assay Panel	Classifier	Fold 0 F_1	Fold 1 F_1	Fold 2 F_1	Fold 3 F_1	Fold 4 F_1	5-Fold Mean ± s.d.	Pooled F_1 (BAcc / ROC-AUC)
Carbon Growth ($n = 2{,}332$)	BacDive-AI RF Baseline	0.8709	0.8720	0.8397	0.7174	0.8646	0.8329 ± 0.0659	0.8448 (0.8842/0.9462)
	DiASPora GBDT	0.8582	0.8819	0.8582	0.7817	0.8238	0.8408 ± 0.0390	0.8483 (0.8879/0.9519)
	Traitar L1-SVM	0.8480	0.8765	0.8727	0.7708	0.8653	0.8467 ± 0.0438	0.8571 (0.8888/0.9430)
	GSPA + DiASPora Hybrid	0.8830	0.9048	0.8824	0.7778	0.8602	0.8616 ± 0.0495	0.8710 (0.9035/0.9567)
Diagnostic Enzymes ($n = 17{,}496$)	BacDive-AI RF Baseline	0.8477	0.8371	0.8490	0.8553	0.8387	0.8456 ± 0.0076	0.8458 (0.8427/0.8962)
	DiASPora GBDT	0.8465	0.8579	0.8560	0.8655	0.8493	0.8550 ± 0.0075	0.8550 (0.8535/0.9061)
	Traitar L1-SVM	0.8620	0.8561	0.8562	0.8714	0.8517	0.8595 ± 0.0076	0.8593 (0.8552/0.9034)
	GSPA + DiASPora Hybrid	0.8655	0.8648	0.8699	0.8842	0.8634	0.8696 ± 0.0085	0.8712 (0.8256/0.9098)

S6 Stage 3 neuro-symbolic tool schemas, multi-seed stability, prospective summary, and extremophile case study

Stage 3 provides five deterministic, read-only genomic and ontological inspection tools to the open-weight instruction model (`run_gspa_agent.py`):

1. `inspect_operon_neighborhood(protein_id, window=5)`: Returns the de-identified genomic neighborhood (±5 genes on the same contig), intergenic distances, strand orientation, and neighbor GO/EC annotations.

2. `inspect_pathway_or_complex_gap(anomaly_id)`: Returns the target pathway or heteromeric complex identifier, fraction of partner steps or subunits present in the genome at $P \geq 0.50$, and list of missing steps or subunits.

3. `query_protein_evidence(protein_id)`: Returns Stage 0 per-channel calibrated probabilities, Pfam/InterPro structural domain annotations, and precomputed RLGC kernel density Bayes factors ($\text{BF}_{\text{operon}}$).

4. `check_symbolic_constraints(go_or_ec_term, taxon_id)`: Queries the deterministic ontology bridge (26,911 direct GO taxon rules and 2,715 GO–Uberon/PO/CL anatomical rules linked to 8,331 taxon constraints across 7,722 anatomical and 1,818 taxonomic hierarchy edges) and returns whether the term is valid in `taxon_id`, along with its nearest valid non-root GO ancestors if forbidden.

5. `propose_resolution(action, target_protein_id, resolved_term, confidence, justification)`: Submits the final structured decision (PROMOTE, RELAX, DEMOTE, or ABSTAIN), which is deterministically verified by the symbolic guardrail before acceptance.

Supplementary Table S14 reports the per-seed and three-seed summary of the full guarded multi-turn Stage 3 configuration across seeds 42, 43, and 44 on the 120-anomaly and 50-unique-protein LRO benchmarks, Supplementary Table S15 reports the complete 17-method prospective CAFA summary (`suite1_cafa6_prospective_summary.tsv`), and Supplementary Table S16 summarizes the five desert extremophile isolates from the Empty Quarter (*Rub' al Khali*).

Table S14. Three-seed evaluation of Stage 3 guarded multi-turn reasoning across the 120-anomaly and 50-gap LRO benchmarks.

Evaluation Run	Random Seed	Precision	Recall	F_1 Score	Accuracy	Violations Accepted	LRO-50 Hit@1 (MRR)
Stage 3 Multi-Turn + Guardrail (120 Anomalies + LRO-50)	Seed 42	0.9787	0.8519	0.9109	0.9250	0/120 (0.0%)	82.00% (0.8740)
Stage 3 Multi-Turn + Guardrail (120 Anomalies + LRO-50)	Seed 43	0.9592	0.8704	0.9126	0.9333	0/120 (0.0%)	84.00% (0.8890)
Stage 3 Multi-Turn + Guardrail (120 Anomalies + LRO-50)	Seed 44	0.9787	0.8519	0.9109	0.9250	0/120 (0.0%)	82.00% (0.8747)
Stage 3 Multi-Turn + Guardrail (3-Seed Mean ± s.d.)	**Seeds 42, 43, 44**	0.9722 ± 0.0092	0.8581 ± 0.0087	0.9115 ± 0.0008	0.9278 ± 0.0039	$0.00 \pm 0.00\%$	$82.67 \pm 0.94\%$ (0.8792 ± 0.0069)

S7 Pan-phylum AllTheBacteria and reference eukaryote analysis

To evaluate whole-genome functional inference across broad prokaryotic and eukaryotic diversity, we processed 73 assembly batches (78,035 extracted assemblies containing 315,671,487 protein sequences) from the AllTheBacteria

Table S15. Complete 17-method prospective CAFA time-based evaluation across 272 post-t_0 targets (`suite1_cafa6_prospective_summary.tsv`).

Method	Targets (n)	MF $F_{\max}$ (f_w)	BP $F_{\max}$ (f_w)	CC $F_{\max}$ (f_w)	Mean $F_{\max}$	Mean f_w	Violations (V)
Naive Term-Frequency Prior (CAFA Baseline 1)	272	0.4156 (0.4074)	0.2305 (0.1865)	0.4430 (0.2813)	0.3631	0.2917	43
InterPro Domain Logistic Regression (InterPro-LR)	272	0.4184 (0.4036)	0.2325 (0.2246)	0.1378 (0.1174)	0.2629	0.2485	76
DeepFRI Graph Convolutional Network	272	0.2790 (0.2729)	0.2568 (0.2247)	0.4666 (0.3669)	0.3341	0.2882	12
eggNOG-mapper v2 Orthology	272	0.4328 (0.4320)	0.2285 (0.2411)	0.4082 (0.3799)	0.3565	0.3510	22
DeepGOCNN (1D-CNN Sequence Model, Retrained)	272	0.4453 (0.4340)	0.2678 (0.2132)	0.5950 (0.4655)	0.4360	0.3709	19
ProteInfer Dilated Residual CNN	272	0.4406 (0.4404)	0.3161 (0.3267)	0.4750 (0.4530)	0.4106	0.4067	2
FoldSeek Structure Alignment (3Di+AA)	272	0.5693 (0.5760)	0.3246 (0.3029)	0.5492 (0.4091)	0.4810	0.4293	48
ESM-2 3B Fine-Tuned PLM	272	0.5714 (0.5526)	0.3641 (0.3310)	0.6609 (0.5426)	0.5321	0.4754	12
ProstT5 Structure-Aware PLM	272	0.5890 (0.5847)	0.3287 (0.2839)	0.6711 (0.5666)	0.5296	0.4784	86
ESM-2 Embedding k-NN (goPredSim, Retrained)	272	0.6136 (0.6144)	0.3499 (0.3048)	0.6417 (0.5194)	0.5351	0.4795	14
DIAMOND BLAST-kNN (CAFA Baseline 2)	272	0.6099 (0.6186)	0.3731 (0.3502)	0.6048 (0.4952)	0.5292	0.4880	27
ESM-2 MLP (Flat BCE, Retrained)	272	0.6441 (0.6320)	0.3192 (0.2894)	0.7045 (0.5928)	0.5559	0.5048	26
DeepGOPlus (1D-CNN + DIAMOND-kNN, Retrained)	272	0.6238 (0.6271)	0.3869 (0.3679)	0.6522 (0.5327)	0.5543	0.5092	22
ESM-2 C-HMCNN (Hierarchical MCLoss, Retrained)	272	0.6502 (0.6378)	0.3360 (0.3037)	0.6961 (0.5886)	0.5608	0.5100	22
GSPA-Seq (Val-Trained Per-Protein Integrator)	272	0.6621 (0.6628)	0.3964 (0.3665)	0.6933 (0.5942)	0.5839	0.5412	644
GSPA Stage 0 (Calibrated Noisy-OR)	272	0.6764 (0.6772)	0.3954 (0.3657)	0.6944 (0.5955)	0.5887	0.5461	160
GSPA Full (Stages 1 + 2A Relaxation)	272	0.6647 (0.6573)	0.3929 (0.3676)	0.6903 (0.5890)	0.5826	0.5380	0

Table S16. Empty Quarter (*Rub' al Khali*) desert extremophile isolate annotation summary across five bacterial genomes.

Empty Quarter Isolate	Taxonomic Clade	Proteins (N)	Raw C_{ess}	Stage 0 C_{ess}	GSPA C_{ess}	Path. Coh. (Raw → GSPA)	pFBA Repairs / Stage 3 Prom.
Salinicola sp. EQ-01	Gammaproteobacteria	3,842	0.8810	0.9120	0.9680	0.4820 → 0.6140	6 repairs / 5 promotions
Halomonas sp. EQ-02	Gammaproteobacteria	4,118	0.8920	0.9210	0.9740	0.4950 → 0.6280	5 repairs / 6 promotions
Pseudomonas sp. EQ-03	Gammaproteobacteria	5,490	0.9140	0.9380	0.9820	0.5190 → 0.6410	4 repairs / 4 promotions
Bacillus endophyticus EQ-04	Bacillota	4,625	0.9010	0.9280	0.9760	0.5040 → 0.6320	5 repairs / 7 promotions
Streptomyces sp. EQ-05	Actinomycetota	7,184	0.8540	0.8960	0.9590	0.4460 → 0.5980	8 repairs / 9 promotions
5-Isolate Mean	**3 Phyla / Classes**	5,051.8	0.8884	0.9190	0.9718	0.4892 → 0.6226	5.6 **repairs** / 6.2 **promotions**

resource (`r0.2` release) paired with reference eukaryotic proteomes. In Stage A, protein sequences were deduplicated by exact SHA-256 amino-acid sequence hashing (73,866,871 unique sequences; 4.27-fold compression) and clustered in linear time via `MMseqs2 linclust` (`-min-seq-id 0.90 -c 0.80 -cov-mode 1`), yielding 36,804,001 non-redundant cluster representatives (8.58-fold total compression). In Stage B, cluster representatives were scored with our multi-modal per-protein stacking integrator GSPA-Seq (combining DIAMOND sequence homology against Swiss-Prot, InterPro/Pfam domain signatures, 16-kernel 1D convolutional DeepGOCNN features, and 2,560-dimensional ESM-2 3B `esm2_t36_3B_UR50D` MLP and hierarchical C-HMCNN representations). In Stage C, representative Stage 0 predictions were mapped back to each chromosome's co-directional operon layout (`make_operons.py`, intergenic distance ≤ 150 bp) and integrated in shared-ontology multi-genome batch mode (`gspa-unified.jar integrate -batch-manifest -enable-priors essentiality,coherence,consistency,anatomical,gap_filling,genomic_context -enforce-stage2 -enforce-consistency-mode relax -lite`).

Supplementary Table S17 reports the cross-phylum breakdown across the 72,533 completed whole-genome integrations (292,133,146 coding sequences across 11,632 species and 37 phyla; complete 37-phylum tables in `results/allthebacteria/insight1_dark_matter_rescue_by_phylum.tsv`, `insight2_complex_stoichiometry_and_operons` `insight3b_phylum_summary.tsv`, `insight4_taxon_anatomical_suppression.tsv`, `insight5_non_orthologous_displacemer` `insight5b_phylum_nogd_vs_divergent.tsv`, and `insight6_genome_size_bifurcation.tsv`). Across all 292,133,146 coding sequences, 119,647,577 (40.96%) are sequence dark matter (pident $< 30\%$ to Swiss-Prot, comprising 102,673,326 zero-alignment orphans [85.81%] and 16,974,251 remote homologs [14.19%]) and 78,843,303 (26.99%) lie in the sequence twilight zone ($30\% \leq$ pident $< 50\%$). Whole-genome GSPA priors rescue 51,375,006 dark-matter proteins (42.94%, of which 39,253,161 [76.41%] are zero-alignment non-orthologous replacements and 12,121,845 [23.59%] are remote divergent homologs) and 60,674,994 twilight-zone proteins (76.96%) above the 0.50 posterior threshold, while suppressing 1,426,276 isolated obligate-complex singleton predictions and eliminating 47,739,228 direct taxonomic (B_3) and 2,388,329,971 propagated anatomical (B_4) cross-kingdom violations (2,436,069,199 total violations suppressed). Supplementary Table S18 details the six genome-size bifurcation regimes across 72,530 bacterial genomes, the stoichiometric rescue versus singleton pruning counts across six obligate prokaryotic complexes, and the partitioning of rescued dark-matter annotations between zero-alignment non-orthologous displacements (pident $= 0.0\%$) and remote divergent orthologs ($0\% <$ pident $< 30\%$).

Table S17. Pan-phylum AllTheBacteria and reference eukaryote whole-genome integration summary across 72,533 complete genomes (292,133,146 coding sequences across 11,632 species and 37 phyla; `insight3b_phylum_summary.tsv`). Major phyla with ≥ 15 genomes and aggregate totals across all 37 phyla are shown. Dark-matter coding sequences have $< 30\%$ pairwise sequence identity to Swiss-Prot; twilight-zone coding sequences have $30\% \leq$ pident $< 50\%$.

Domain	Phylum	Genomes	Species	Total ORFs	Dark ORFs (%)	Dark Rescued (%)	Twilight Rescued (%)	B_6 Operon Prom.	B_2 Single. Supp.	$B_3 + B_4$ Viol. Supp.
Bacteria	Actinomycetota	31,304	2,212	135,787,329	49,575,316 (36.51%)	25,447,143 (51.33%)	23,288,398 (78.36%)	387,047,892	620,895	1,109,581,418
Bacteria	Pseudomonadota	20,734	4,952	86,910,368	35,130,850 (40.42%)	12,314,404 (35.05%)	19,200,043 (74.57%)	275,556,172	525,406	730,708,905
Bacteria	Bacillota	12,675	2,591	38,427,622	18,751,008 (48.80%)	7,496,485 (39.98%)	11,582,066 (78.79%)	126,964,510	126,663	332,708,913
Bacteria	Bacteroidota	3,341	1,101	13,846,301	8,929,800 (64.49%)	3,002,893 (33.63%)	3,110,686 (75.01%)	31,426,851	61,081	118,687,904
Bacteria	Unclassified Bacteria	3,256	1	13,334,981	5,073,736 (38.05%)	2,749,156 (54.18%)	2,539,869 (79.34%)	42,908,989	68,378	125,488,234
Bacteria	Campylobacterota	327	176	613,728	316,688 (51.60%)	70,868 (22.38%)	179,071 (76.43%)	1,389,408	6,557	3,388,042
Bacteria	Spirochaetota	216	116	686,593	417,557 (60.82%)	79,303 (18.99%)	164,138 (72.27%)	1,200,460	2,573	3,557,425
Bacteria	Cyanobacteriota	115	88	636,008	387,725 (60.96%)	66,609 (17.18%)	123,930 (64.43%)	772,478	2,131	3,900,664
Bacteria	Desulfobacterota_I	72	51	238,413	122,590 (51.42%)	20,517 (16.74%)	68,747 (73.56%)	490,493	2,890	1,054,475
Bacteria	Fusobacteriota	67	33	156,125	84,893 (54.38%)	27,478 (32.37%)	45,289 (78.60%)	447,651	493	1,057,866
Bacteria	Deinococcota	65	44	240,057	127,706 (53.20%)	12,068 (9.45%)	66,125 (71.14%)	363,159	1,037	943,560
Bacteria	Desulfobacterota	51	47	183,894	94,154 (51.20%)	12,147 (12.90%)	52,038 (72.92%)	368,935	2,417	738,804
Bacteria	Verrucomicrobiota	48	36	178,665	108,910 (60.96%)	13,284 (12.20%)	39,798 (68.31%)	250,875	837	716,049
Bacteria	Chlamydiota	39	15	52,103	28,715 (55.11%)	3,168 (11.03%)	13,755 (73.23%)	76,019	185	246,605
Bacteria	Planctomycetota	31	22	179,627	119,699 (66.64%)	7,844 (6.55%)	32,924 (63.90%)	148,586	720	548,839
Bacteria	Fibrobacterota	28	17	83,173	52,000 (62.52%)	4,491 (8.64%)	18,083 (70.53%)	93,768	620	275,522
Bacteria	Acidobacteriota	27	25	129,097	75,601 (58.56%)	9,525 (12.60%)	31,271 (70.82%)	169,672	716	510,527
Bacteria	Thermotogota	23	17	48,238	24,366 (50.51%)	5,365 (22.02%)	15,387 (76.00%)	114,880	258	231,840
Bacteria	Myxococcota	18	13	129,765	77,482 (59.71%)	6,920 (8.93%)	29,166 (65.85%)	174,271	395	479,641
Bacteria	Bacteroidota_A	15	11	44,235	23,386 (52.87%)	4,031 (17.24%)	12,434 (73.70%)	86,754	289	220,605
Archaea / Euk. / 17 Minor Phyla	17 Phyla	81	64	226,824	119,401 (52.64%)	20,307 (17.01%)	60,779 (73.24%)	401,904	1,735	1,023,355
Total / Pan-Phylum Summary	**37 Phyla**	72,533	11,632	292,133,146	119,647,577 (40.96%)	51,375,006 (42.94%)	60,674,994 (76.96%)	870,453,527	1,426,276	2,436,069,199

Table S18. Genome-size bifurcation across 72,530 bacterial genomes (`insight6_genome_size_bifurcation.tsv`), obligate complex stoichiometric selection (`insight2_complex_stoichiometry_and_operons.tsv`), and non-orthologous gene displacement (pident $= 0.0\%$) versus remote sequence divergence ($0\% <$ pident $< 30\%$) across macromolecular complexes and metabolic pathways (`insight5_non_orthologous_displacement.tsv`).

Analysis Stratum / Category	Subgroup / Complex / Pathway Slot	Genomes / Phyla	Species / GO Term	Primary Count (n_1)	Secondary Count (n_2)	Dark / Total (%)	Twilight / Rate	Key Metric / Ratio
Part A: Genome-Size Bifurcation ($n = 72{,}530$ Genomes) (n_1: Mean ORFs; n_2: Mean B_6/Genome; Dark: Mean Dark %; Twilight: Mean Twi. %; Key Metric: B_5 Gaps/Genome [per 1k ORFs])	$<$ 1,500 ORFs (Streamlined / Host-Associated)	413 (17p)	238 species	1,152.3	2,269.0 (1.914/ORF)	50.62%	34.55%	0.14 (0.108/1k)
	1,500–2,499 ORFs (Small Free-Living / Host-Adapted)	5,114 (26p)	2,018 species	1,985.6	5,624.6 (2.863/ORF)	45.83%	36.82%	0.70 (0.351/1k)
	2,500–3,499 ORFs (Compact Generalists)	15,283 (25p)	2,539 species	2,982.9	10,139.7 (3.416/**ORF**)	47.62%	36.31%	1.59 (0.518/1k)
	3,500–4,499 ORFs (Modal Bacterial Generalists)	36,243 (20p)	3,123 species	4,084.8	12,038.3 (2.950/ORF)	37.46%	22.83%	3.29 (0.819/1k)
	4,500–5,499 ORFs (Versatile Environmental)	9,323 (14p)	2,072 species	5,061.7	16,532.0 (3.248/ORF)	39.44%	25.12%	12.70 (2.484/1k)
	$\geq$ 5,500 ORFs (Large Environmental / Secondary Metab.)	6,154 (17p)	2,549 species	6,609.9	15,494.6 (2.431/ORF)	46.57%	31.72%	19.10 (2.876/**1k**; 136.4×)
Part B: Stoichiometric Selection Across 6 Obligate Complexes (n_1: Coherence-Rescued Subunits; n_2: Pruned Singleton Hallucinations)	70S Ribosome (30S + 50S Structural Subunits)	37 phyla	`GO:0005840`	6,339,928	15	100.00% rescued	190,778 dark	422,662 : 1 rescue/prune
	DNA-Directed RNA Polymerase Holoenzyme	36 phyla	`GO:0000428`	4,733,571	197	99.996% rescued	373,272 dark	24,028 : 1 rescue/prune
	Proton-Transporting F_1F_0 ATP Synthase	35 phyla	`GO:0045259`	1,215,204	108	99.991% rescued	541,889 dark	11,252 : 1 rescue/prune
	Heterotetrameric DNA Gyrase ($GyrA_2B_2$)	35 phyla	`GO:0009330`	192,355	2,659	98.64% rescued	2,527 dark	72.3 : 1 rescue/prune
	Respiratory Complex I (NuoA–N vs. Soluble N-Module)	34 phyla	`GO:0005747`	25,179	1,202,957	97.95% pruned	857 dark	47.8 : 1 prune/rescue
	SecYEG Protein Export Translocon Complex	30 phyla	`GO:0031522`	104	220,519	99.95% pruned	0 dark	2,120 : 1 prune/rescue
Part C: Non-Orthologous Displacement vs. Remote Sequence Divergence (n_1: Zero-Alignment pident = 0%; n_2: Remote Divergent 0 $<$ pident $<$ 30%; Dark: Total Dark Rescued; Key: % NOGD)	Thiol / Mycothiol S-Transferase Pathway Gap-Fill (B_5)	4 phyla	`GO:0004364`	22,183	20	22,203	194 twi.	99.91% **NOGD**
	ATP Synthase F_0 Peripheral Membrane Subunits (B_2)	20 phyla	`GO:0045259`	520,762	11,373	532,135	117,388 twi.	97.86% **NOGD**
	RNA Polymerase Small Accessory Subunits ω/δ (B_2)	24 phyla	`GO:0000428`	292,899	18,737	311,636	110,214 twi.	93.99% **NOGD**
	All Co-Directional Operon Synteny Promotions (B_6)	37 phyla	All Operon Slots	258,892,062	54,001,319	312,893,381	216,828,547 twi.	82.74% **NOGD**
	Adenylate Cyclase Signaling Module (`KEGG:M00859`, B_2)	27 phyla	`GO:0004016`	559,705	128,365	688,070	384,699 twi.	81.34% **NOGD**
	Cell-Wall LPS Hexosyltransferase (`KEGG:M00888`, B_2)	36 phyla	`GO:0016758`	1,053,238	475,574	1,528,812	1,334,895 twi.	68.89% **NOGD**
	Xenobiotic Glycosyltransferase Pathway Gap-Fill (B_5)	17 phyla	`GO:0015020`	7,299	13,352	20,651	38,403 twi.	35.34% NOGD
	DNA Gyrase Catalytic Core (GyrA/B, B_2)	15 phyla	`GO:0006265`	23	2,308	2,331	111,537 twi.	1.00% NOGD (99.00% Remote)
	RNA Polymerase Catalytic Core Transcription (B_2)	27 phyla	`GO:0006351`	158	33,280	33,438	1,264,790 twi.	0.47% NOGD (99.53% Remote)
	ATP Synthase Rotary Catalysis & Carbonyl Reductase (B_2/B_5)	25 phyla	`GO:0046933/0004090`	3	6,640	6,643	135,606 twi.	0.05% NOGD (99.95% Remote)